10.4.2 if-else statement

In the above examples of **if**, you have seen that, a block of statements are excecuted only if the condition evaluates to true. What if there is another course of action to be followed if the condition evaluates to false. There is another form of **if** that allows for this kind of either or condition by providing an else clause. The syntax of the if-else statement is given below:

```
if ( expression)
{
        True-block;
}
else
{
        False-block;
}
Statement-x
```

In if-else statement, first the expression or condition is evaluated to either true of false. If the result is true, then the statements inside true-block is executed and false-block is skipped. If the result is false, then the statement inside the false-block is executed i.e., the true-block is skipped.

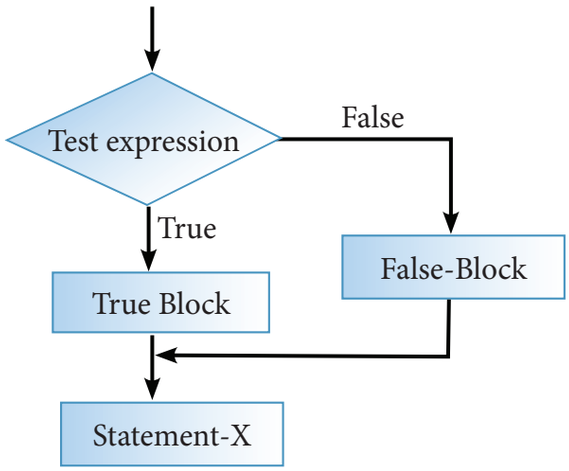

Illustration 10.2 C++ program to find whether the given number is even number or odd number using if-else statement

```
#include <iostream>
using namespace std;
int main()
{
        int num, rem;
        cout<< "\n Enter a number: ";
        cin>>num;
        rem = num % 2;
        if (rem==0)
                cout<< "\n The given number" <<num<< " is Even";
        else
                cout<< "\n The given number "<<num<< " is Odd";
        return 0;
}
```

Output

```
Enter number: 10
The given number 10 is Even
```

In the above program, the remainder of the given number is stored in rem. If the value of rem is zero, the given number is inferred as an even number otherwise, it is inferred as on odd number.

11th Std Computer Science_EM Chapter 10.indd 155 06-01-2023 15:09:57